NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Mid-Term Examination I: *April'2025*
Subject: *Data Structures [CSPC100/ ITPC100/ AIPC100]*
Branches: *CSE, IT, AI&ML* ***(Regular)***

MM: *15*
Duration: *50 mins*

NOTE: 1. *The question paper contains* ***NINE*** *questions.*
2. *All questions are compulsory.*
3. *Assume suitable data, if missing.*

Q1. Consider **implicit array representation** for storing binary trees. Find the **minimum size** of an array so that one can store any binary tree with ***n*** vertices.

Q2. Consider the following statements with respect to an **almost(nearly) complete binary tree:-**

S1: There is only a single almost complete binary tree with n nodes.
S2: There are two distinct almost complete binary trees with n leaves.

Choose any one correct option from the following.
(i) Both are true **(ii)** Both are false **(iii)** S1 is true, S2 is false **(iv)** S1 is false, S2 is true
(v) S1 is true, S2 may or may not true **(vi)** S1 may or may not true, S2 is true

Q3. Match each entry in column-A with a unique entry in column-B.

	Column-A	Column-B
(i)	Pre-order traversal	(a) Symmetric order
(ii)	In-order traversal	(b) En-order
(iii)	Post-order traversal	(c) level-order
		(d) Depth-first

Q4.
(i) Is it possible to have a graph(undirected) with the degree sequence: (1, 3, 3, 3, 5, 6, 6)
(ii) Is it possible to have a simple graph with the degree sequence: (1, 1, 3, 3, 3, 4, 6, 7)

Q5. The pre-order and post-order traversal of a **full binary tree** are as follows:

Pre-order: {1, 2, 4, 8, 9, 5, 3, 6, 7}
Post-order: 8, 9, 4, 5, 2, 6, 7, 3, 1}

From the above-given information, if possible, then construct the tree; otherwise, explain the reason for non-construction.

Q6. A ***strict k-ary tree*** is one in which every node has **0** or **k** children. If **x** is the number of internal nodes of the ***strict k-ary tree***, then find the number of leaves in it.

Q7. Consider a stack S of integers, and a function f is defined on different states on the stack as follows:

$f(\emptyset) = 0$; initially when the stack is empty, otherwise
$f(push(S, i)) = \max(f(S), 0) + i$; for all stacks S and integers i.

If a stack S contains the integers 2, -3, 2, -1, 2 in order from bottom to top, find the value of $f(S)$.

Q8. The following infix expression is converted to postfix using a stack.

A + B * C $ D – E * F

Find the top two elements of the stack after scanning the entire symbols of the input string. Here, the symbol **$** represents an exponential operator, and other symbols have usual meanings.

Q9. Write a program (in C/C++ or pseudo code) to insert at the front of a **singly linked list** whose head node's pointer to the pointer is denoted by ***list,*** and the node structure is as follows:

```
struct Node{
    int value;
    struct Node* next;
}
```

Further, assume the following **method header** for the method to insert at the front of the singly linked list:

```
void insertAtFront(struct Node ** list, int x)
```

Marks Distribution

Q.1 to 4: 1 Mark
Q.5 to 8: 2 Marks
Q.9: 3 Marks